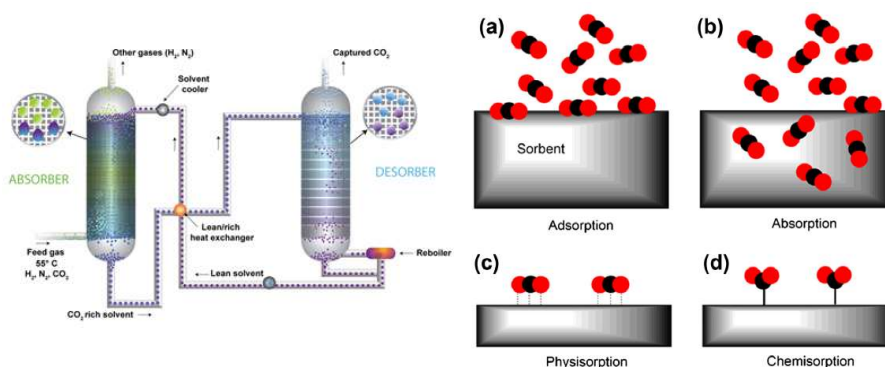


Carbon Capture Methods



Department of Chemical Engineering
National Institute of Technology, Warangal

8/14/2025

CH447 – CO₂ Capture, Sequestration & Utilizations

Green House Gases



- Energy, agriculture, automobile, industry and wastes from domestic and industrial uses are biggest sources of green house gases (GHGs).
- Primary focus is to capture the gases rather than limiting the usage as we are more energy and services dependent.

Concentration Ranges of Carbon Dioxide (as Volume Percentages) from a Variety of Sources and Possible Contaminants

Source	CO ₂ Concentrations (%)	Typical Impurities
Power generation (post-combustion)	10–15	N ₂ , H ₂ O, SO _x , NO _x
Steel making (blast furnace gas)	18–20	N ₂ , SO _x , NO _x , O ₂
Cement production	c. 100	N ₂ , O ₂
Fermentation	c. 100	H ₂ O, H ₂ S
Natural gas streams	0–8	N ₂ , O ₂ , H ₂ S, C ₁ –C ₄ hydrocarbons
Natural deposits	90–100	N ₂ , O ₂ , He
Atmosphere	0.04	N ₂ , O ₂ , SO _x , NO _x , particulates

8/14/2025

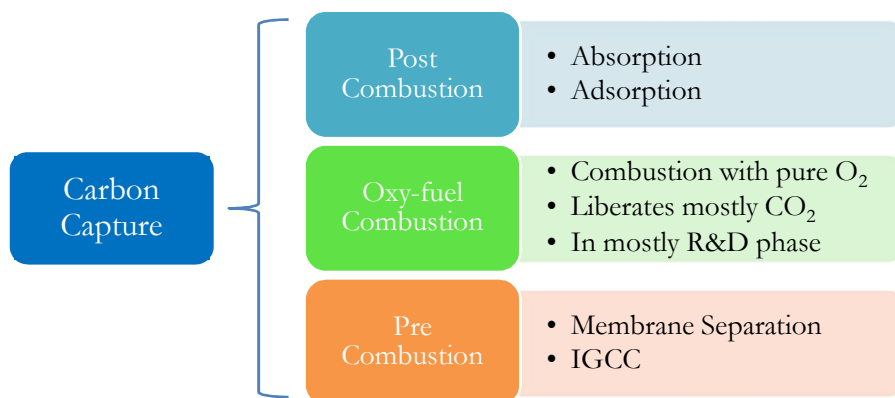
CH447 – CO₂ Capture, Sequestration & Utilizations

2

Green House Gases



- Energy, agriculture, automobile, industry and wastes from domestic and industrial uses are biggest sources of green house gases (GHGs).
- Primary focus is to capture the gases rather than limiting the usage as we are more energy and services dependent.



8/14/2025

CH447 – CO_2 Capture, Sequestration & Utilizations

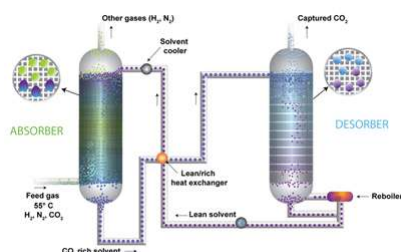
3

Post-Combustion Capture

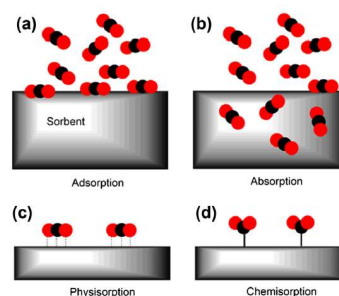


- Capture of CO_2 from the exhaust flue gas of combustion process is known as post-combustion capture.
- Chemical and/or physical absorption, physical adsorption and membrane separation are available.
- Flue gas : N_2 (80 %), CO_2 (10 – 15%), O_2 (3 – 4%) and traces of SO_2 , NO_x and CO, H_2S , fly ash, acid and water vapours.

Absorption and Adsorption



<http://chemicalfiles.blogspot.com/2012/05/absorption.html>



8/14/2025

CH447 – CO_2 Capture, Sequestration & Utilizations

4

Post-Combustion Capture



- Physical absorption – CO_2 is soluble in the absorbing liquid (water, amines) and no reaction happens with solvent.
 - Chilled methanol – Rectisol Process
 - N-methyl-2-pyrrolidine – Purisol Process
 - Dimethyl ether of polyethylene glycol – Selexol Process
 - Propylene carbonate – Flour Process
- Solvents show good loading capacity, easy regeneration, applicability at high partial pressure of CO_2 .
- Physical absorption solvents are expensive leading to chemical solvents usage when gas volumes are huge.
- Chemical absorption – CO_2 reacts with a component in liquid phase to form a product.
 - Aqueous alkanolamine
 - Hot potassium carbonate

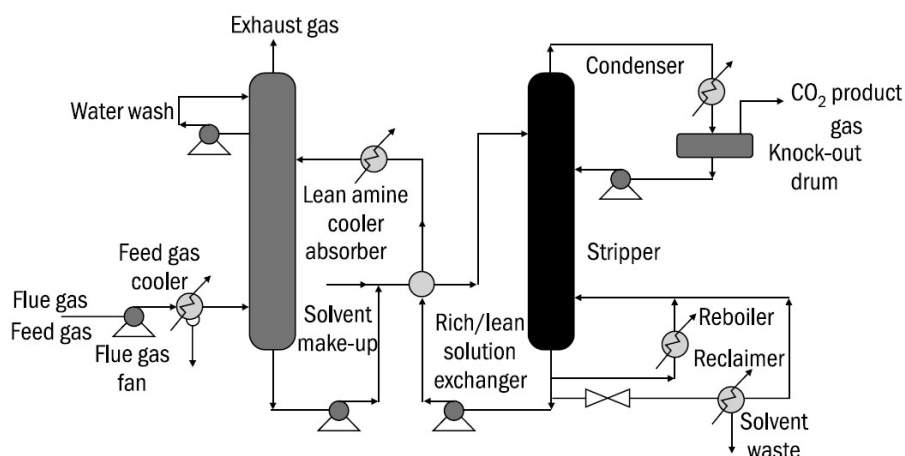
Physical
absorbents

8/14/2025

CH447 – CO_2 Capture, Sequestration & Utilizations

5

Post-Combustion Capture



Typical schematic diagram for CO_2 capture by chemical absorption

8/14/2025

CH447 – CO_2 Capture, Sequestration & Utilizations

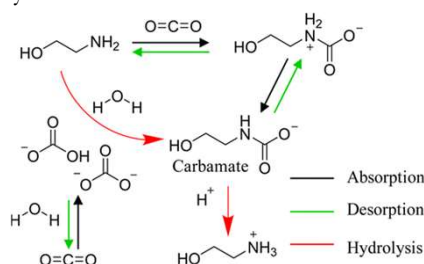
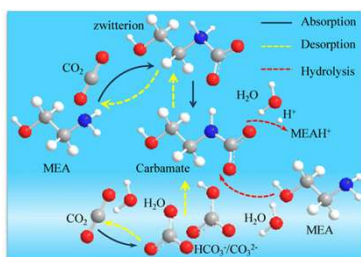
6

Post-Combustion Capture



Aqueous alkanolamine solutions for CO₂ capture:

- Known for adaptability, ability to remove till low concentrations of CO₂, Lower cost
 - Monoethanolamine (MEA) - primary
 - Diethanolamine (DEA) – secondary
 - Methyldiethanolamine (MDEA) – tertiary
- Primary and secondary amines in water can instantaneously form carbamate ions which increases CO₂ solubility.



<https://pubs.acs.org/doi/10.1021/acs.est.5b02356>

8/14/2025

CH447 – CO₂ Capture, Sequestration & Utilizations

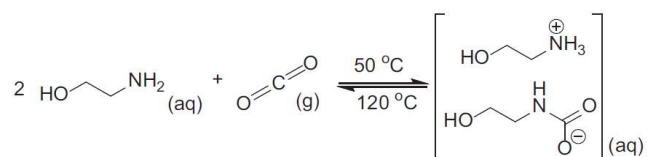
7

Post-Combustion Capture



Aqueous alkanolamine solutions for CO₂ capture:

- MEA is dissolved in water (~ 30% by weight) – Absorption occurs at 50 °C
- The adsorption selectivity is 28:1 for CO₂ to N₂ using this primary amine.



Carbon dioxide absorption by MEA in aqueous solution.

- Because of chemical bond formation, there is energy requirement for the bond breaking (desorption or regeneration process)
- This occurs typically at 120 °C which requires a large sum of energy.
- MEA undergoes chemical degradation under these conditions. Fresh solvent must be added periodically. This will increase operating costs.
- For every tonne of CO₂ captured – 2 kg of MEA is replaced.

8/14/2025

CH447 – CO₂ Capture, Sequestration & Utilizations

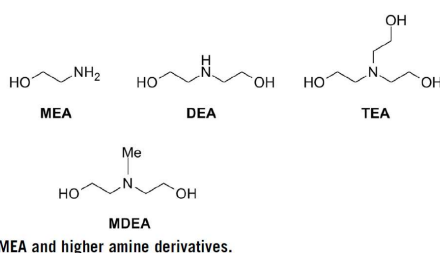
8

Post-Combustion Capture



Aqueous alkanolamine solutions for CO₂ capture:

- For every 100 tonne CO₂ captured per day – 220 kg MEA/ day replacement
- 80 tonnes MEA /year which costs around 11 million euros/ year.
- Decomposition of MEA – complex molecules with N₂ that are more harmful than CO₂ to atmosphere.
- Alcohols have hydrogen bond interactions which will effect sorption process.
- Molecules with high hydroxyl groups have high activities rather than selectivity's.



8/14/2025

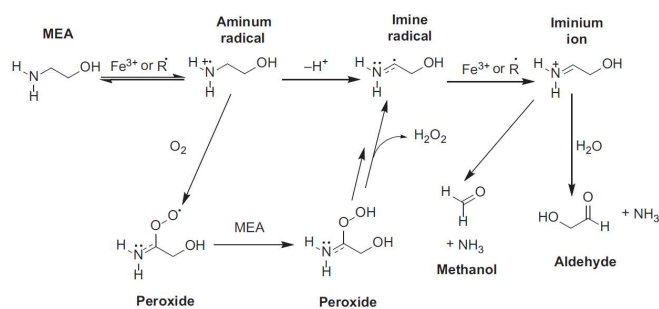
CH447 – CO₂ Capture, Sequestration & Utilizations

9

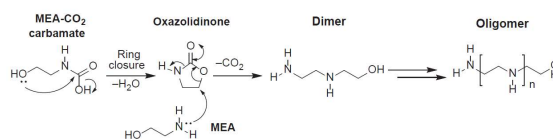
Post-Combustion Capture



Aqueous alkanolamine solutions for CO₂ capture:



An oxidative MEA degradation pathway showing the intermediate steps in the formation of ammonia and aldehydes.



An example of a carbamate polymerisation mechanism occurring in the stripper

8/14/2025

CH447 – CO₂ Capture, Sequestration & Utilizations

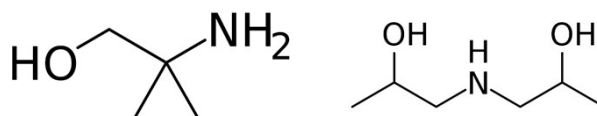
10

Post-Combustion Capture



Aqueous alkanolamine solutions for CO₂ capture:

- Disadvantage of limited loading as the reaction should happen between two MEA and one CO₂ molecules.
- Tertiary amines do not react directly but increases the hydrolysis of CO₂ with a reaction stoichiometry of 1:1. The loading will be more and regeneration cost is lower.
- Amine methyl propanol (AMP), Di-isopropanolamine have high reversible reaction with CO₂ giving easy regeneration costs.



- Low corrosion, stable solvents, less steam consumption are advantages of above hindered amines.
- Blended amines – primary/secondary + tertiary/hindered amines

8/14/2025

CH447 – CO₂ Capture, Sequestration & Utilizations

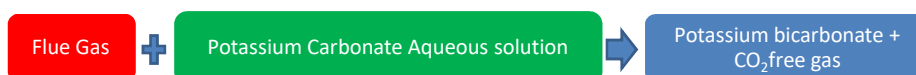
11

Post-Combustion Capture



Carbonate Process

- Developed in 1950 – 70 requires high partial pressure of CO₂ (217 kPa)



- The reaction rate is increased with increased temperature. Upon heating the liquid is regenerated.
- The process also separates H₂S gas from flue gas as it reacts with potassium carbonate.
- Applications
 - CO₂ removal from NH₃ plants
 - Iron ore reductions
 - Natural gas treatments and recycling gas purification
- High partial pressure requirement, large heat requirements for regeneration of solvent, foam and corrosion are disadvantages.

8/14/2025

CH447 – CO₂ Capture, Sequestration & Utilizations

12

Post-Combustion Capture



Chilled NH₃ Process

- Flue gas is cooled between temperatures (0 – 10 °C) – Reduces the volume of gas and increases CO₂ concentration.
- This increases CO₂ capture efficiency in the absorber with low NH₃ emission.
- Regeneration is carried out at ~120 °C and 20 bar – high pressure of CO₂ with low concentrations of moisture and NH₃.
- Advantages
 - Less expensive solvents
 - Low partial pressure CO₂ can be treated
 - Less energy requirement
- Disadvantages
 - Corrosion problems
 - Complex scheme compared to MEA

8/14/2025

CH447 – CO₂ Capture, Sequestration & Utilizations

13

Post-Combustion Capture



CO₂ capture from power plants

- Flue gas : N₂ (80 %), CO₂ (10 – 15%), O₂ (3 – 4%) and traces of SO₂ , NO_x and CO, H₂S, fly ash, acid and water vapours.
- Monoethanolamine – ability to absorb CO₂ in low concentration. Used this to capture CO₂ from power plant flue gas.

Energy requirements for various CO₂ removal schemes

Scheme	Coal requirements (kg/kWh)
No CO ₂ removal	0.35
Amine scrubbing	0.66–1.68
Cryogenic fractionation	0.78–7.04
Membrane separation	0.72–1.41

- In case of Indian coal based flue gas the cost will increase much.
- More oxygen in flue gas can make process corrosive and also the loss of MEA by degradation.

8/14/2025

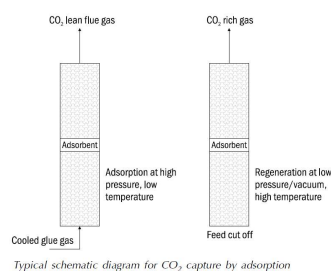
CH447 – CO₂ Capture, Sequestration & Utilizations

14

Post-Combustion Capture - Adsorption



- **Adsorption** – solid/fluid interaction. This is utilized in solid catalyzed fluid reactions and gas separations.
- The gas particles held on to the pores of adsorbent surface by means van der Waals' forces.
- The phenomenon is reversible, rapid with low heat of adsorption.
- Adsorption rate = $f(T, \text{gas concentrations, surface forces, pore sizes/surface area})$
- Separation by
 - ✓ Steric effect – structure difference
 - ✓ Equilibrium effect – saturation loading
 - ✓ Kinetic effect - diffusion properties in the pores of adsorbent
- High selectivity and high capacity
- Large surface area



8/14/2025

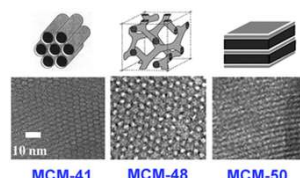
CH447 – CO₂ Capture, Sequestration & Utilizations

15

Post-Combustion Capture - Adsorption



- Common adsorbents – microporous materials with high surface areas and specific sites of adsorption. Ex: Zeolites, Mesoporous silica (MCM-41).
- Functionalization of the pores can enhance the CO₂ separation.
- Mesoporous Silica
 - ✓ Uniform large pores, tunable size
 - ✓ High surface areas
 - ✓ Distributed active sites
 - ✓ Hydroxyl groups
- Activated carbon, clays, silica gel, activated alumina – commercial adsorbents



MCM-41 MCM-48 MCM-50
Charles et al. Chem. Soc. Rev., 2013,42, 3663-3670

Typical properties of AC

Property	Micropore	Mesopore	Macropore
Diameter, Å	<20	20–500	>500
Pore volume (cm ³ /g)	0.15–0.5	0.02–0.1	0.2–0.5
Surface area (m ² /g)	100–1000	10–100	0.5–2
Particle density: 0.6–0.9 g/cm ³ ; porosity: 0.4–0.6			

8/14/2025

CH447 – CO₂ Capture, Sequestration & Utilizations

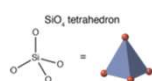
16

Post-Combustion Capture - Adsorption

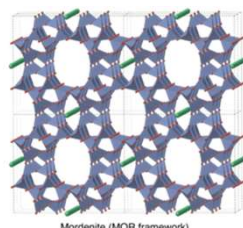
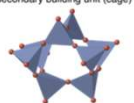


CO₂ Capture from flue gas by adsorption

- Natural materials – passing flue gas through unmined coals.
- Synthetic adsorbents – Activated carbon, zeolites etc
- Adsorption is a slow process. It cannot handle large volumes of flue gas and large concentrations of CO₂.



secondary building unit (cage)



Zeolite formation

Natural and synthetic zeolites

<https://en.wikipedia.org/wiki/Zeolite>

8/14/2025

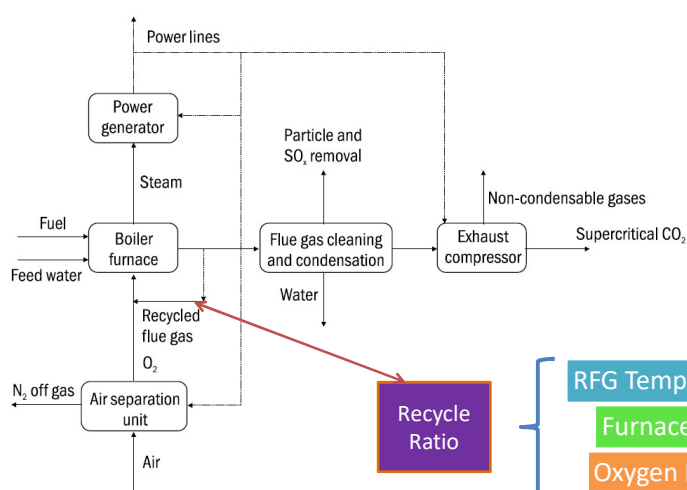
CH447 – CO₂ Capture, Sequestration & Utilizations

17

Oxy-Fuel Combustion



- Fuel is fired with an oxygen-rich gas (~ 95% O₂) .



8/14/2025

CH447 – CO₂ Capture, Sequestration & Utilizations

18

Oxy-Fuel Combustion



Gas composition for bituminous coal ($\text{CH}_{1.1}$, $\text{O}_{0.2}$, $\text{N}_{0.017}$, and $\text{S}_{0.015}$) fired with air and oxygen [25]

Gas	Air firing	Oxy-firing
CO_2	17%	64% by volume
H_2O	8.9%	34%
NO_x	2,770f ppm	10,700f ppm
SO_x	2,470 ppm	9,400 ppm
Moles	1	0.26

f is fractional conversion of coal nitrogen to NO_x

➤ Advantages

- Production of CO_2 rich flue gas – can be sequestered easily
- Mass and volume of flue gas reduction (~75%)
- Process equipment size also reduced drastically
- Minimization of heat losses, less NO_x production (feed is free from N_2)

➤ Disadvantages

- High flame temperature because of pure O_2 – mixing of recycle flue gas with O_2 can reduce the damage on process equipment.

8/14/2025

CH447 – CO_2 Capture, Sequestration & Utilizations

19

Oxy-Fuel Combustion



- Reduction of NO_x or SO_2 is not fully understood but can be estimated from empirical equations.
- Okazaki and Liu proposed the following equation

$$(\text{NO})_{\text{exit}} = \frac{\alpha F_N}{R \eta} + 1$$

α – fuel fraction converted into nitric oxide, F_N – fuel nitrogen

R – Recycle ratio, η – Fraction of NO destroyed while re-burning

Percentage reduction of NO_x with recycle ratio of RFG

Recycle ratio of RFG	% reduction of NO_x
0	No reduction
2.25	69.5
2.75	78.5
3.25	82.5

8/14/2025

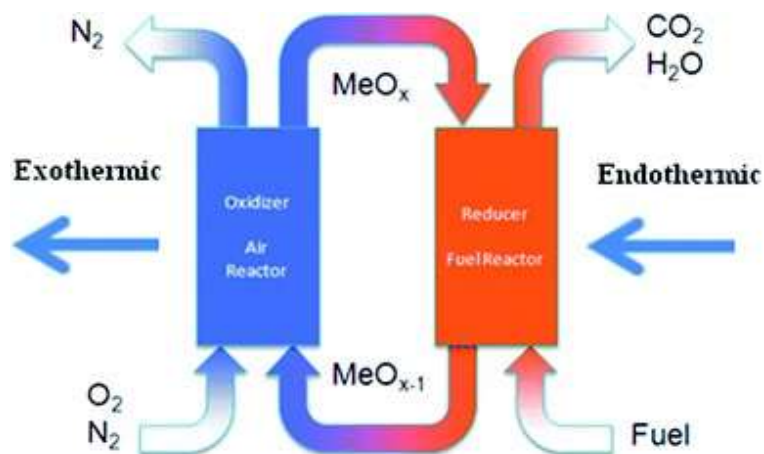
CH447 – CO_2 Capture, Sequestration & Utilizations

20

Chemical Looping Combustion (CLC)



- Two fluidized beds are employed.



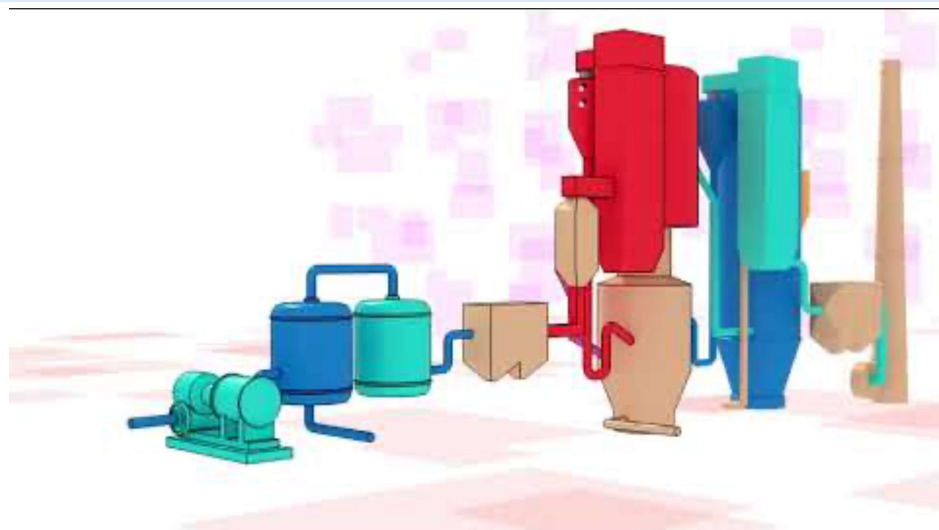
<https://doi.org/10.1039/C6FD00033A>

8/14/2025

CH447 – CO₂ Capture, Sequestration & Utilizations

21

Chemical Looping Combustion (CLC)



Source : <https://www.youtube.com/watch?v=DwTY4vBvR0c>

8/14/2025

CH447 – CO₂ Capture, Sequestration & Utilizations

22

Chemical Looping Combustion (CLC)



- The metal reacts with air to form metal oxide – **oxidizer bed**
- The metal oxide is mixed with fuel and oxygen – fuel is oxidized and metal oxide is **reduced** to metal.
- The reduced metal is sent back to the oxidizer for reuse.
- This technique is in laboratory stage only as capture technology.
- **Challenges**
 - Identification of **suitable metal** for several cycles without performance loss and sustainable
 - Effective **separation of metal and ash** (in case of solid fuel)
- **Advantages**
 - The combustion efficiency increases by selecting the oxygen carriers appropriately.
 - Flue gas is usually free from NO_x as there is no nitrogen in air sample.

8/14/2025

CH447 – CO₂ Capture, Sequestration & Utilizations

23

Pre-Combustion Capture (CLC)

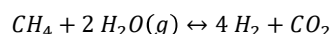
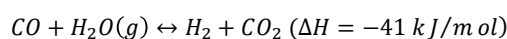
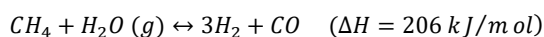


Pre-combustion capture

- Hydrogen Membrane Reforming (HMR)
- Sorber Enhanced Water-Gas-Shift Reaction (SEWGR)
- Integrated Gasification Combined Cycle (IGCC)

Hydrogen Membrane Reforming

- This is also referred as membrane water-gas-shift reaction (MWGSR) – steam reforming + water-gas-shift reaction



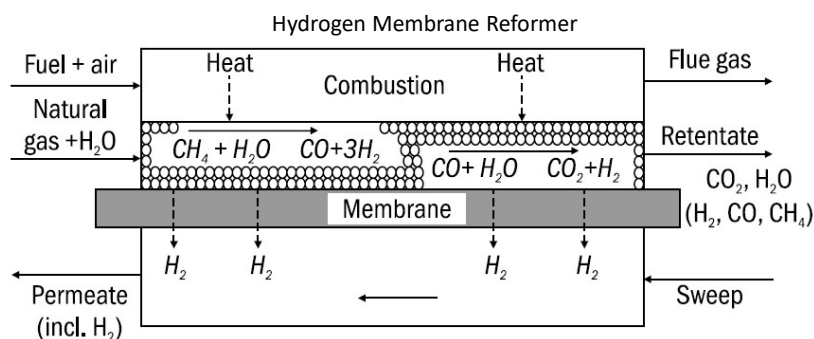
- Membrane acts as a separation with permeate side of H₂ gas and downstream with unreacted CH₄ and water.
- Pressure – 36 bar, Temperature – 400 to 600 °C ideal for HMR operation

8/14/2025

CH447 – CO₂ Capture, Sequestration & Utilizations

24

Pre-Combustion Capture (CLC)



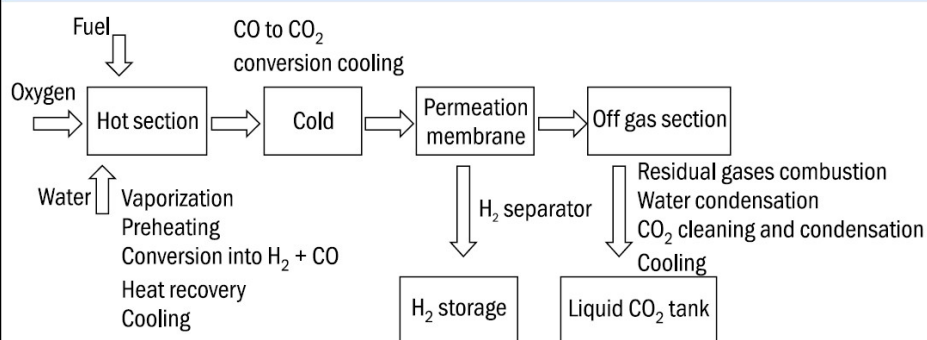
- Reforming and water gas shift reaction happens parallel. Heat is generated through combustion of fuel.
- The generated H_2 is used as source of fuel which is free from carbon.
- Development of membranes with high temperature operations are important. **Palladium-ceramic** composite membrane is selective to H_2 and temperature resistant.

8/14/2025

CH447 – CO₂ Capture, Sequestration & Utilizations

25

Pre-Combustion Capture (CLC)



Pd- ceramic membrane module

- Major Challenges in membrane synthesis
 - Thin layer membrane synthesis (3 -9 μm)
 - Low cost production
 - Removal of metal supporting foils

8/14/2025

CH447 – CO₂ Capture, Sequestration & Utilizations

26

Pre-Combustion Capture (CLC)



Sorption Enhance Water-Gas-Shift Reaction

- High temperature water-gas-shift reaction (HTS) produces H_2 and CO_2 around $400\text{ }^\circ\text{C}$ which is **CO equilibrium reaction**.
- The gas is **cooled** to $200\text{ }^\circ\text{C}$ to get higher conversion of CO to CO_2 and further captured using amine based solvents.
- The remaining gas (H_2) is heated to $400\text{ }^\circ\text{C}$ before feeding into gas turbines. The on and off temperature cycles makes this process **expensive**.

SEWGS = HTS + High temperature CO_2 adsorption

- This shifts the equilibrium towards products which forms maximized H_2 .
- Regenerated CO_2 from adsorbent material is pure which can be sequestered directly without further processing.
- Carbon capture project (Phase-II) – CCP2 are developing on these technologies.

8/14/2025

CH447 – CO_2 Capture, Sequestration & Utilizations

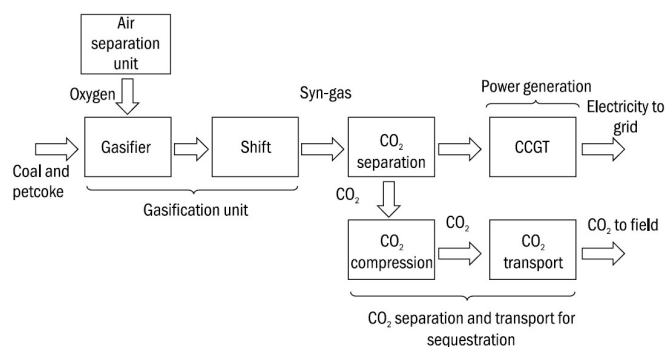
27

Pre-Combustion Capture (CLC)



Integrated Gasification Combined Cycle

- This technology utilizes the gasification of low value fuels into hydrogen rich synthesis gas by integration of a gasification unit.
- Heating value of syngas – $4.2 - 4.6\text{ MJ/N m}^3$.
- Impurities such as Sulphur compounds, NH_3 , metals, alkylates, ash and particulates are removed before feeding into turbine.



8/14/2025

CH447 – CO_2 Capture, Sequestration & Utilizations

28

Pre-Combustion Capture (CLC)



Integrated Gasification Combined Cycle

➤ Advantages

- ✓ Low solvent requirement – Less capital cost
- ✓ 90% Hg removal
- ✓ 18% less NO_x than supercritical pulverized coal as fuel
- ✓ 50% less SO_x than supercritical pulverized coal as fuel
- ✓ 42% less particulate matter than supercritical pulverized coal as fuel
- ✓ Inorganic components are trapped in an ash
- ✓ 30% less water consumption than conventional coal technology
- ✓ Produces half solid waste than conventional technology
- ✓ This is particularly useful for coals with high Sulphur content

8/14/2025

CH447 – CO₂ Capture, Sequestration & Utilizations

29

Gas Processing Before CO₂ capture



- Flue gas from a power plant – SO₂, NO_x, O₂, N₂, H₂O, and CO₂.
- SO₂ + MEA → corrosive salts (irreversible reaction)
- SO₂ scrubber is needed before feeding the flue gas into amine based solvents.
- Limestone and active alkali metal neutralizing agents processing to remove Sulphur traces.
- SO₃ + H₂O → H₂SO₄ (aerosols in wet scrubbers)
- NO_x can be produced by fuel combustion or thermal NO_x from environment N₂. Catalytic reduction and flame temperature control can reduce the NO.
- Tri-mer TRI-NO_x Multistage scrubbing – HCl, SO₂, NO_x, Cl₂, HNO₃, HF etc.

8/14/2025

CH447 – CO₂ Capture, Sequestration & Utilizations

30

Future Perspectives



- Absorption/adsorption materials research is key in post-combustion capture.
- The focus is more on solid materials having high porosity which makes the diffusion of gases easy when compared to viscous liquids/ liquid solvents.
- Solids can be used without any dilution. They have relatively low vapour pressures than liquids and solutions which reduces the evaporation losses.
- Ionic liquids, Polymers, Metal Organic Frameworks are having above characteristics to explore.
 - Ionic liquids – low vapour pressures at ambient conditions.
 - MOFs – High tenability in terms of pore sizes
 - Polymers – Easy to synthesize and lower costs.

8/14/2025

CH447 – CO₂ Capture, Sequestration & Utilizations

31

Future Perspectives - Ionic Liquids

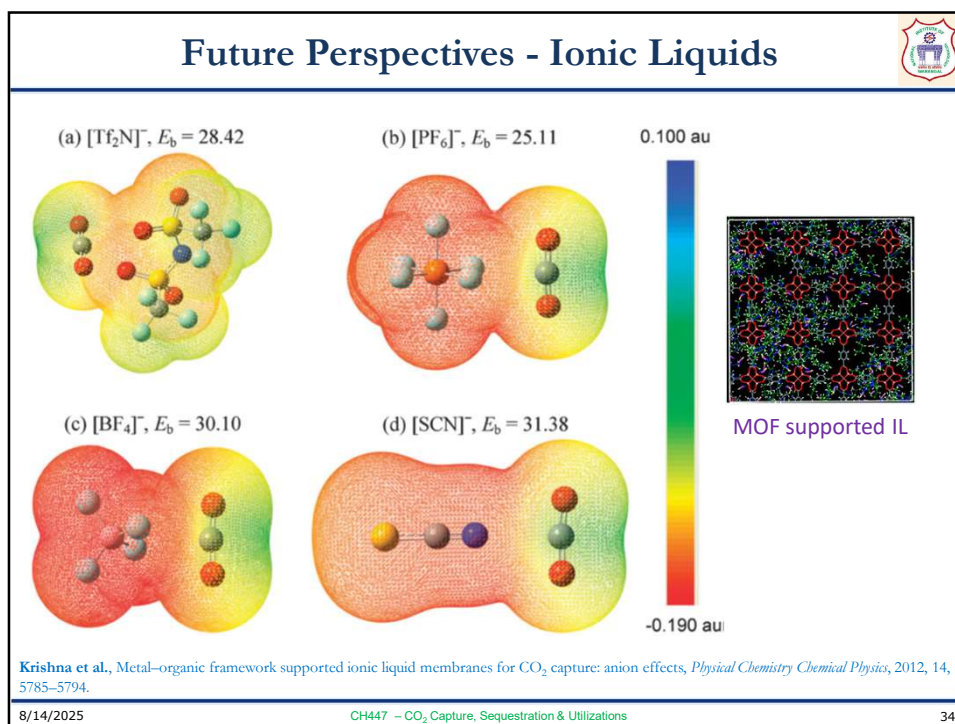
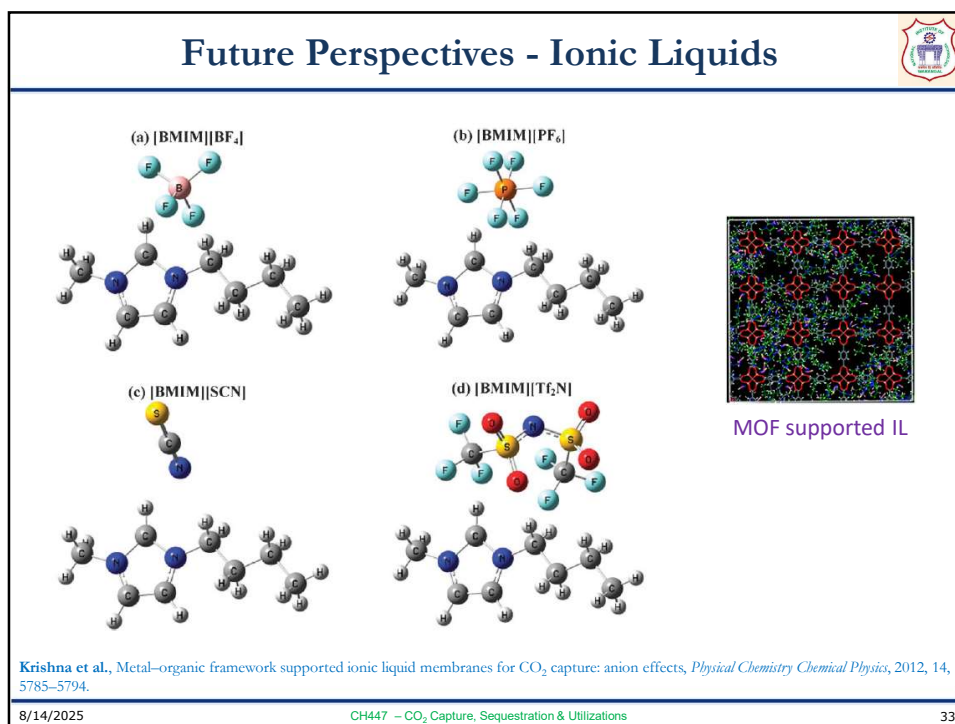


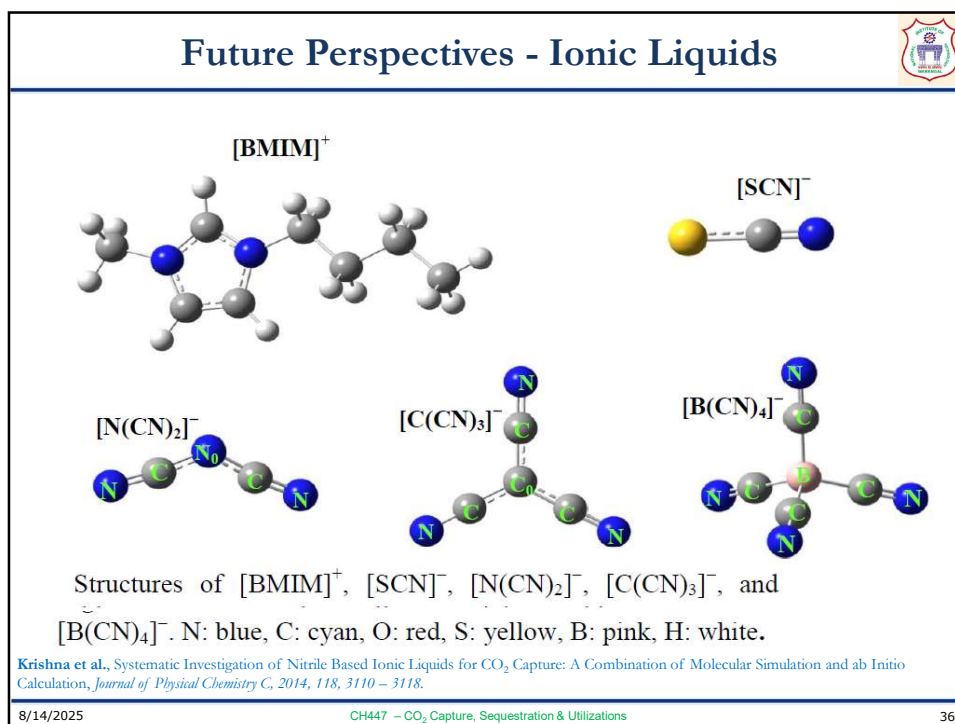
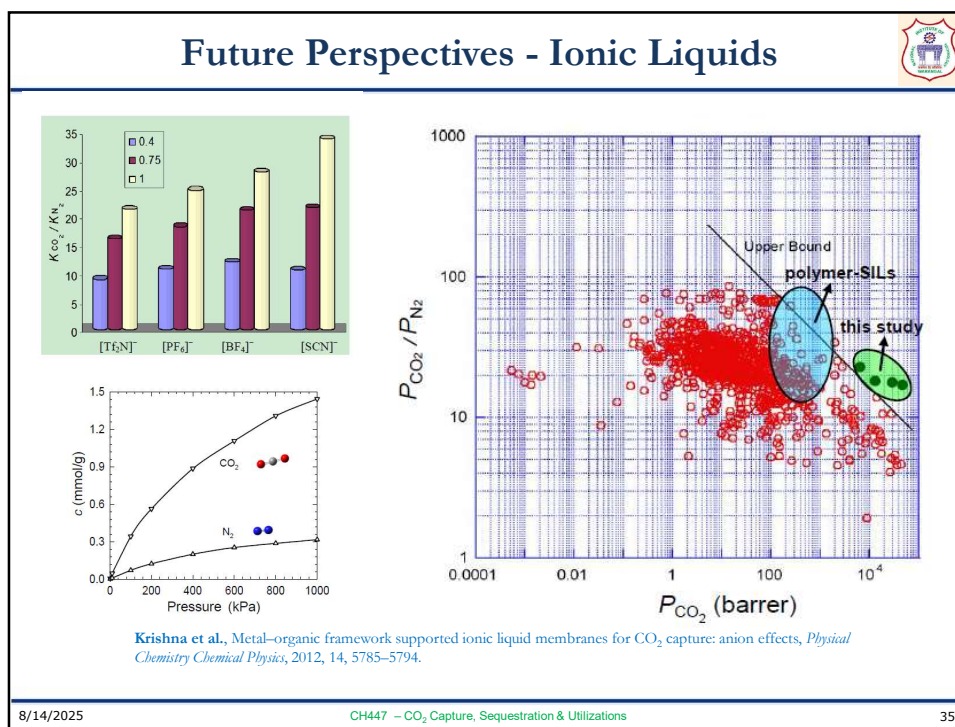
- Ionic liquids (ILs) have emerged as a unique class of materials with melting temperatures lower than 100 °C or substantially lower than normal salts.
- The structure is composed of ions (cations and anions)
- Common cations are bulky and asymmetric (organic) in nature, such as imidazolium, pyridinium, pyrrolidinium, ammonium etc.
- Anions may range from simple halides, inorganic ions to large organic ions, for example, [BF₄]⁻, [PF₆]⁻, [Ac]⁻, [Tf₂N]⁻, [CF₃COO]⁻ etc.
- ✓ Unique characteristics distinguishing ILs from conventional solvents are
 - ✓ wide range of melting temperature (-40 to 400 °C),
 - ✓ high thermal stability (up to 400 °C),
 - ✓ low vapor pressure, weakly coordinating properties
 - ✓ low flammability, high conductivity (both ionic and thermal)
 - ✓ broad electrochemical potential window (-4 to 4V)

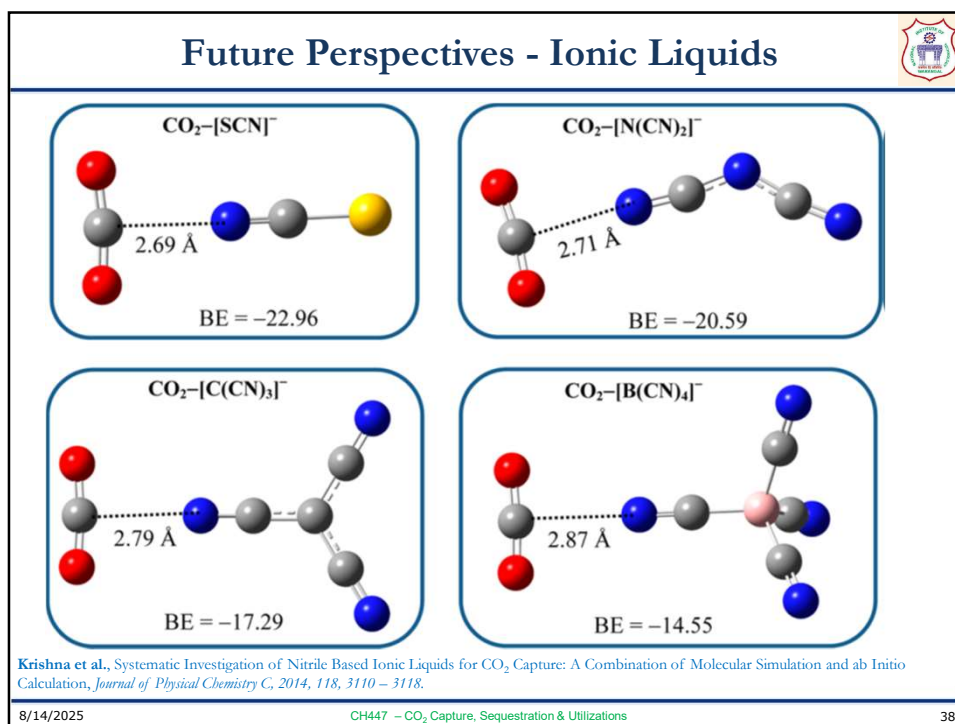
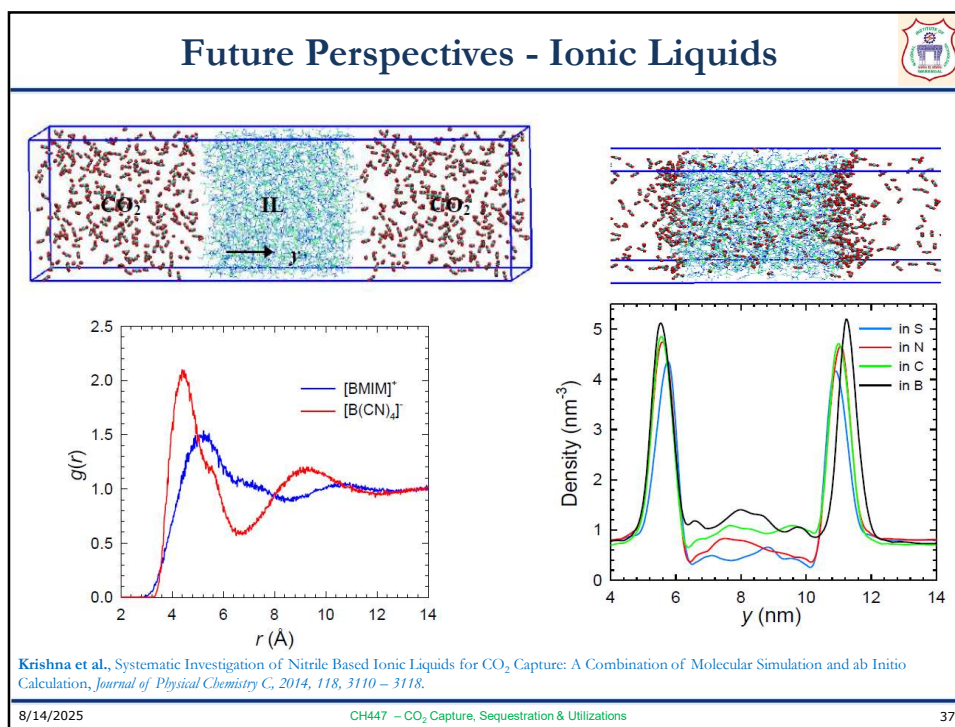
8/14/2025

CH447 – CO₂ Capture, Sequestration & Utilizations

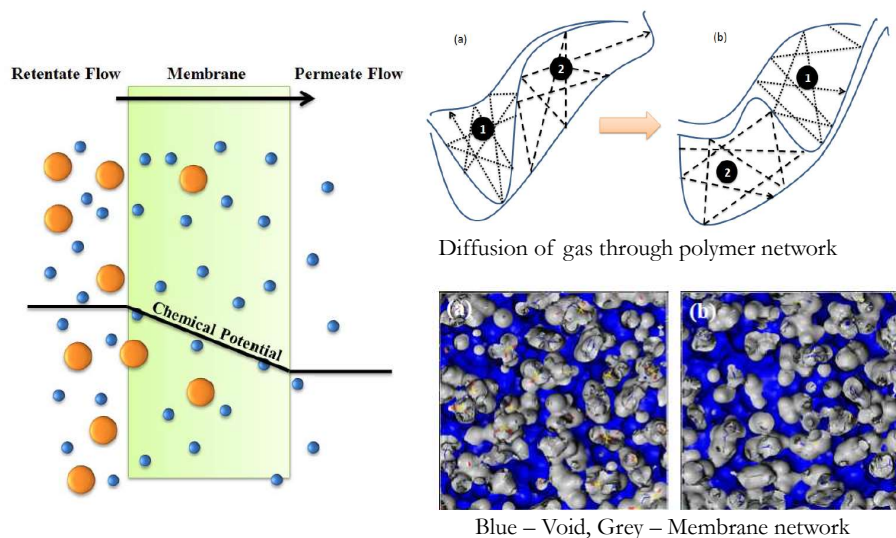
32







Future Perspectives - Polymers



Weiji et al., CO₂ capture in poly(ionic liquid) membranes: atomistic insight into the role of anions, *Physical Chemistry Chemical Physics*, 2013, 15, 651-658.

8/14/2025

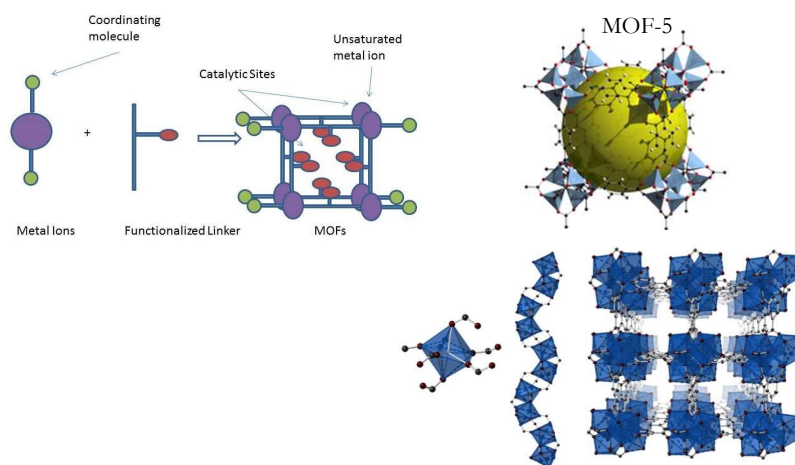
CH447 – CO₂ Capture, Sequestration & Utilizations

39

Future Perspectives – MOFs



- Metal–organic frameworks (MOFs) are a class of compounds consisting of metal ions or clusters coordinated to organic ligands to form 1D, 2D, 3D structures.



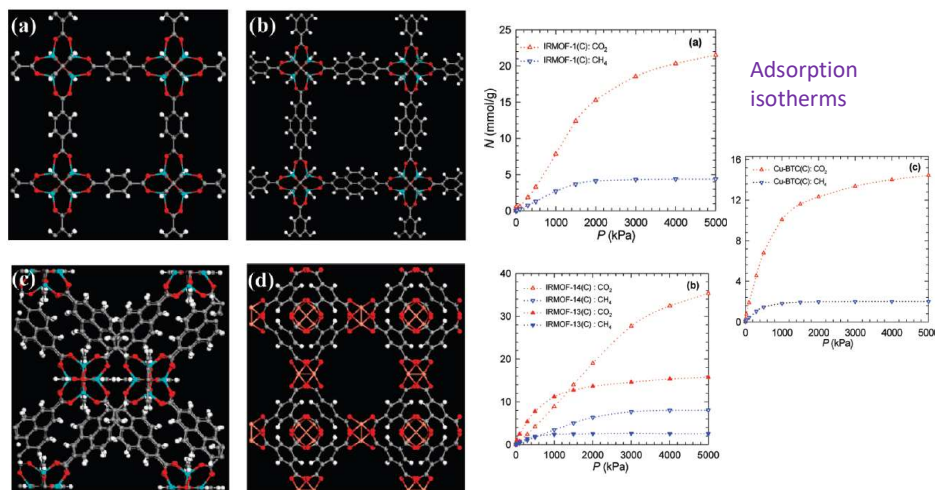
<https://commons.wikimedia.org/wiki>

8/14/2025

CH447 – CO₂ Capture, Sequestration & Utilizations

40

Future Perspectives – MOFs



Atomic structures of (a) IRMOF-1, (b) IRMOF-14, (c) IRMOF-13, (d) Cu-BTC

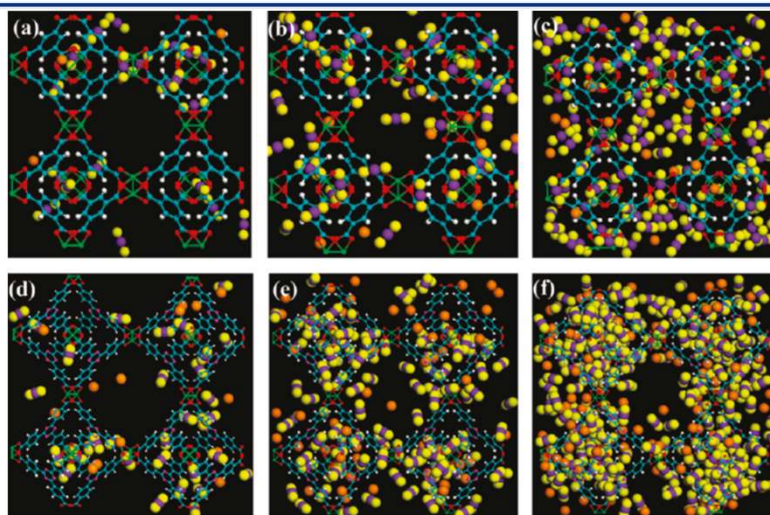
Babarao et al. Molecular Simulations for Adsorptive Separation of CO₂/CH₄ Mixture in Metal-Exposed, Catenated, and Charged Metal–Organic Frameworks, *Langmuir* **2009**, 25(9), 5239–5247.

8/14/2025

CH447 – CO₂ Capture, Sequestration & Utilizations

41

Future Perspectives – MOFs



Simulation snapshots of the CO₂/CH₄ mixture at pressures of (a) 100, (b) 300, and (c) 1000 kPa in Cu-BTC (top) and (d) 300, (e) 1000, and (f) 3000 kPa in PCN-6' (bottom): Cu, green; O, red; C, cyan; N, pink; H, white; CH₄, orange; and CO₂, purple for C and yellow for O.

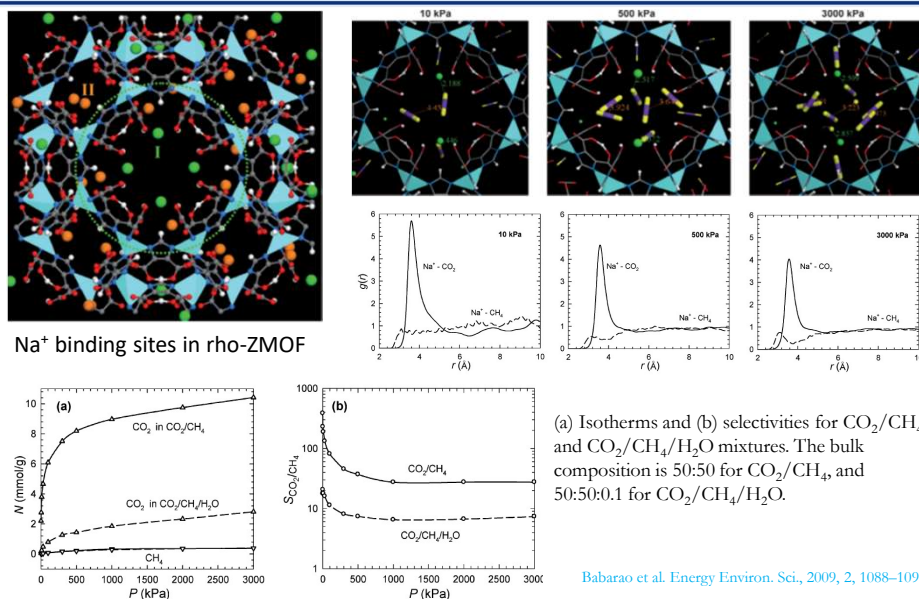
Babarao et al. Molecular Simulations for Adsorptive Separation of CO₂/CH₄ Mixture in Metal-Exposed, Catenated, and Charged Metal–Organic Frameworks, *Langmuir* **2009**, 25(9), 5239–5247.

8/14/2025

CH447 – CO₂ Capture, Sequestration & Utilizations

42

Future Perspectives – MOFs

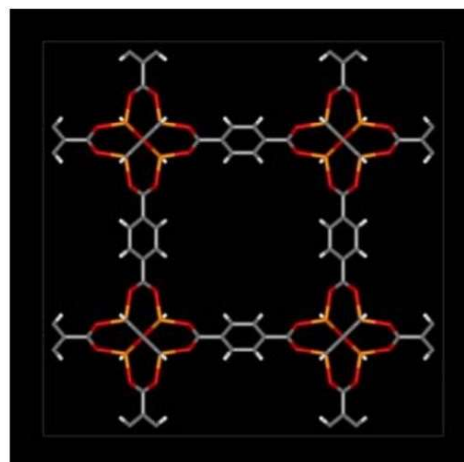
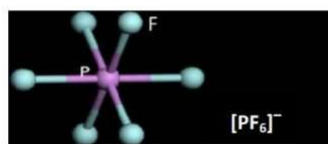
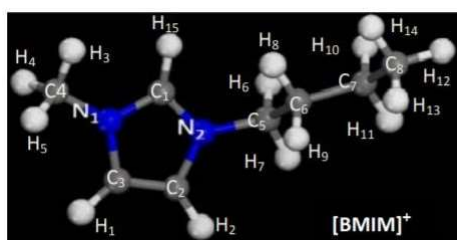


8/14/2025

CH447 – CO₂ Capture, Sequestration & Utilizations

43

IL/MOF Composite for CO₂ Capture



IRMOF-1

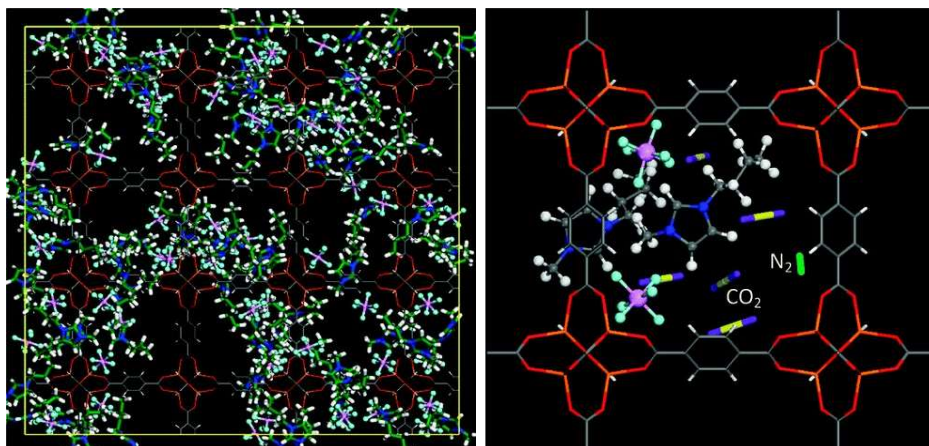
Chen et al. Ionic Liquid/Metal–Organic Framework Composite for CO₂ Capture: A Computational Investigation, *J. Phys. Chem. C*, 2011, 115, 44, 21736–21742.

8/14/2025

CH447 – CO₂ Capture, Sequestration & Utilizations

44

IL/MOF Composite for CO₂ Capture



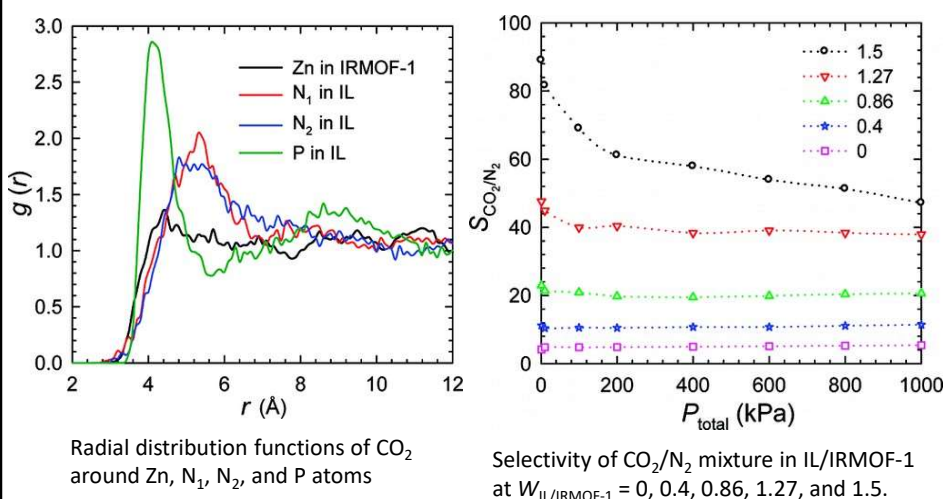
Chen et al. Ionic Liquid/Metal–Organic Framework Composite for CO₂ Capture: A Computational Investigation, *J. Phys. Chem. C*, 2011, 115, 44, 21736–21742.

8/14/2025

CH447 – CO₂ Capture, Sequestration & Utilizations

45

IL/MOF Composite for CO₂ Capture

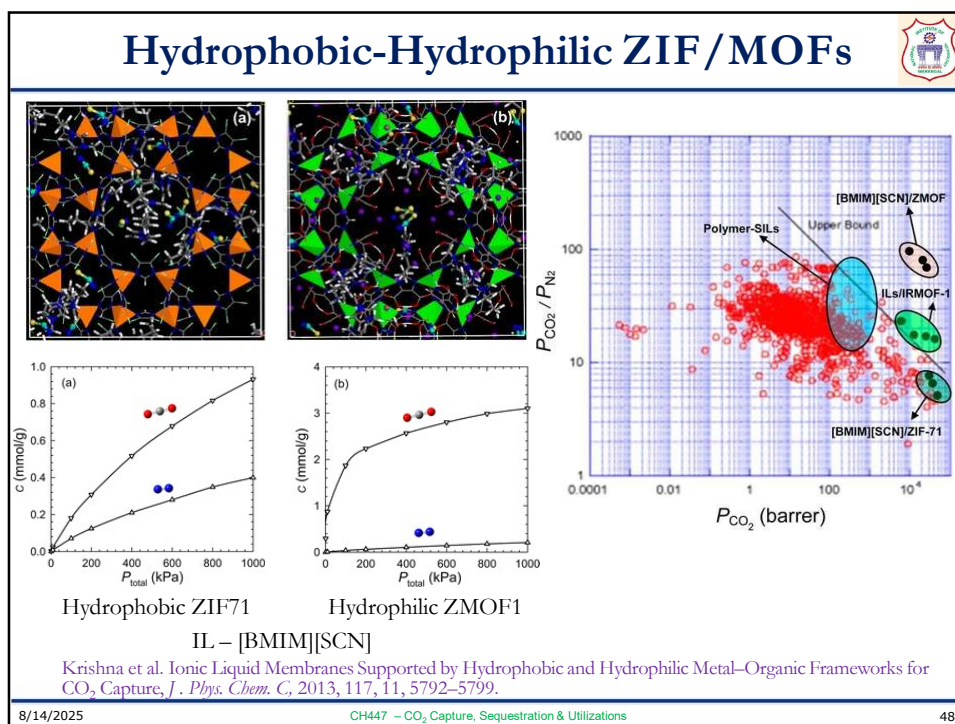
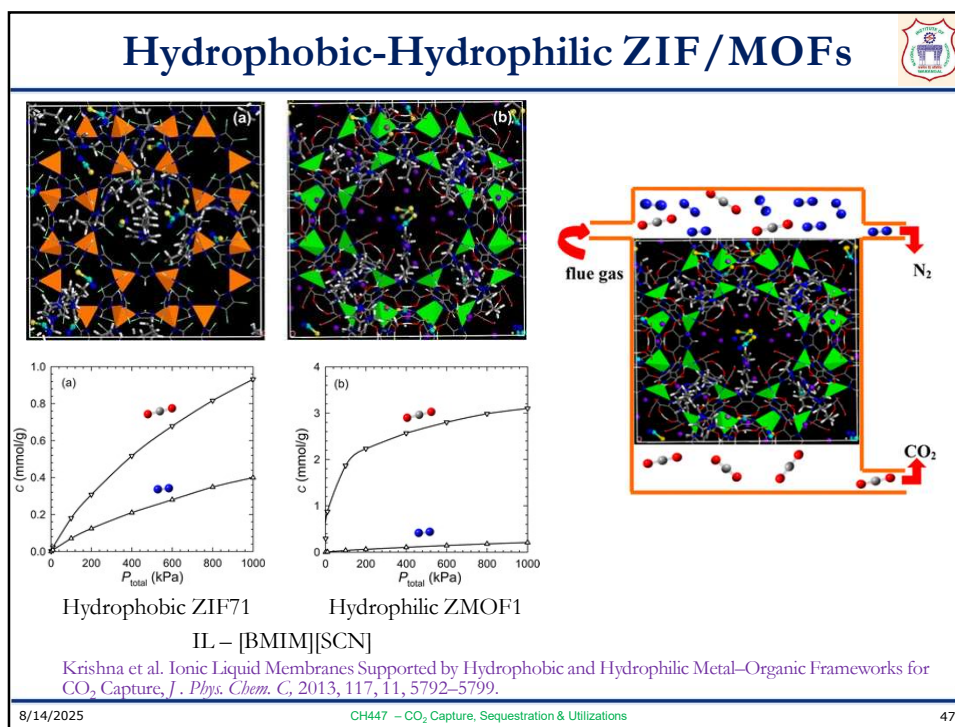


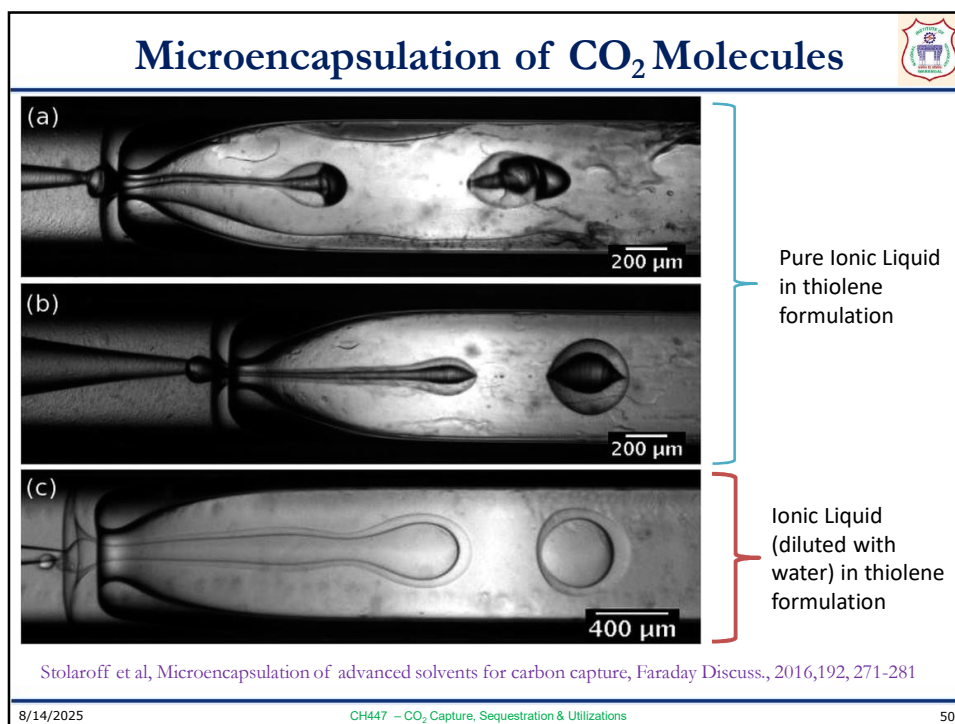
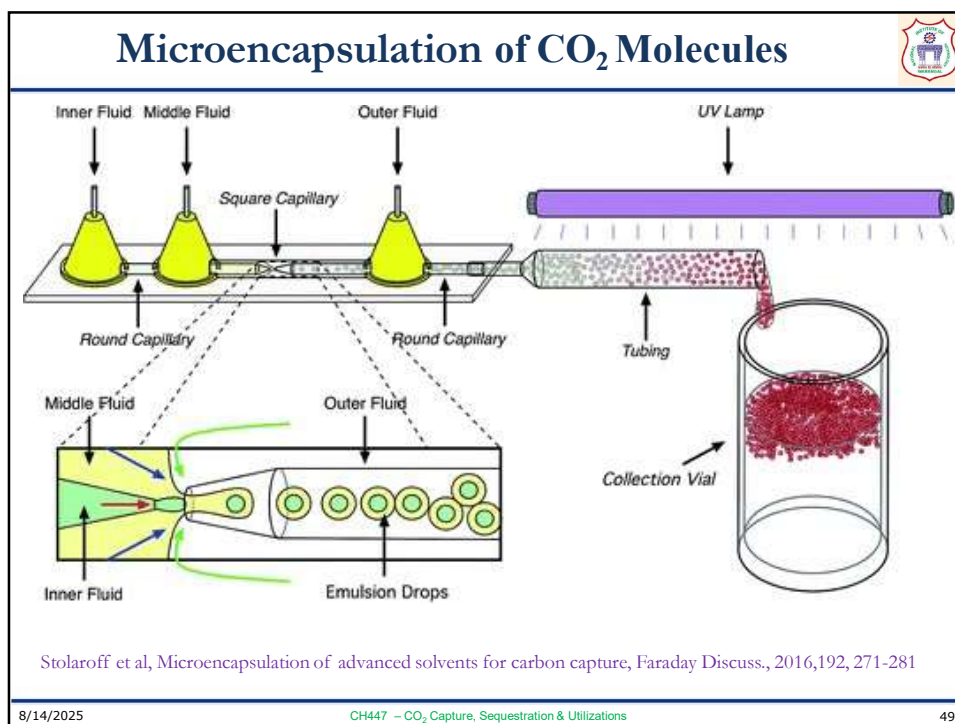
Chen et al. Ionic Liquid/Metal–Organic Framework Composite for CO₂ Capture: A Computational Investigation, *J. Phys. Chem. C*, 2011, 115, 44, 21736–21742.

8/14/2025

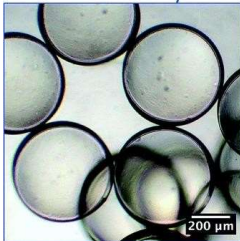
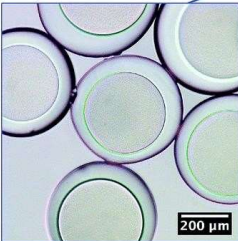
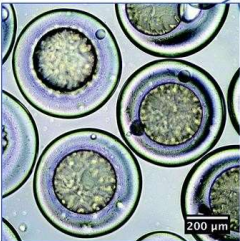
CH447 – CO₂ Capture, Sequestration & Utilizations

46





Microencapsulation of CO ₂ Molecules											
	Koech-anol	Koech-anol/water	DBU/Hexanol	P ⁶⁶⁶¹⁴ Inda	P ⁶⁶⁶¹⁴ 2CN	NDIL0231	NDIL0231/water	NDIL0230	NDIL023/water	P ²²²² Bnlm (solid)	P ²²²² Bnlm / water
Semiosil						×					
Thiol-ene		✓				×					✓
SiTRIS							✓				

Good properties for encapsulation

Marginal properties for encapsulation

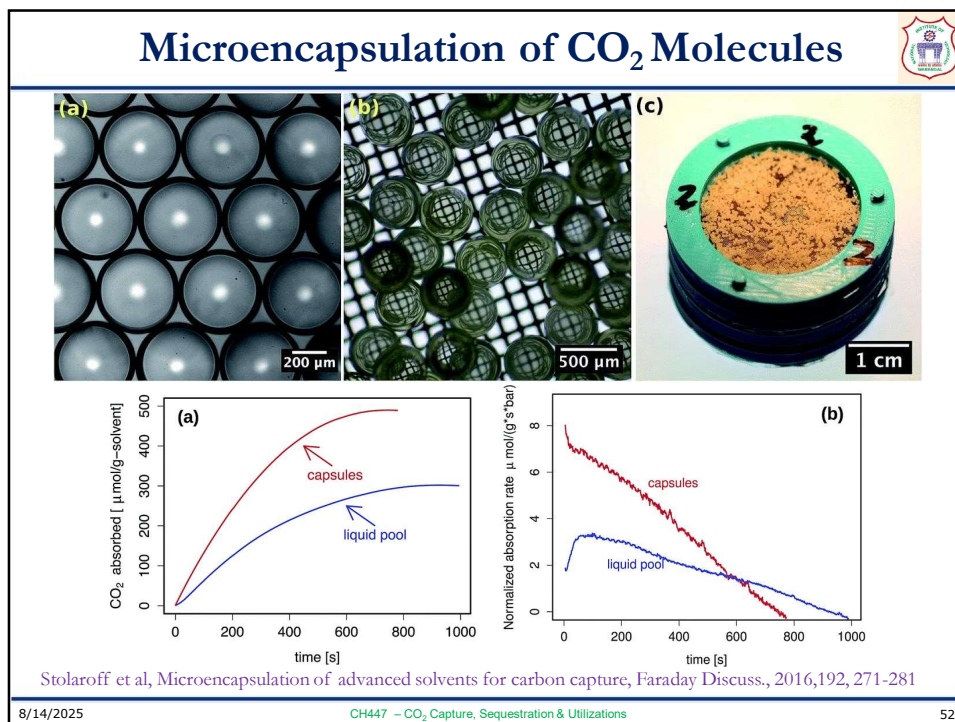
Not compatible

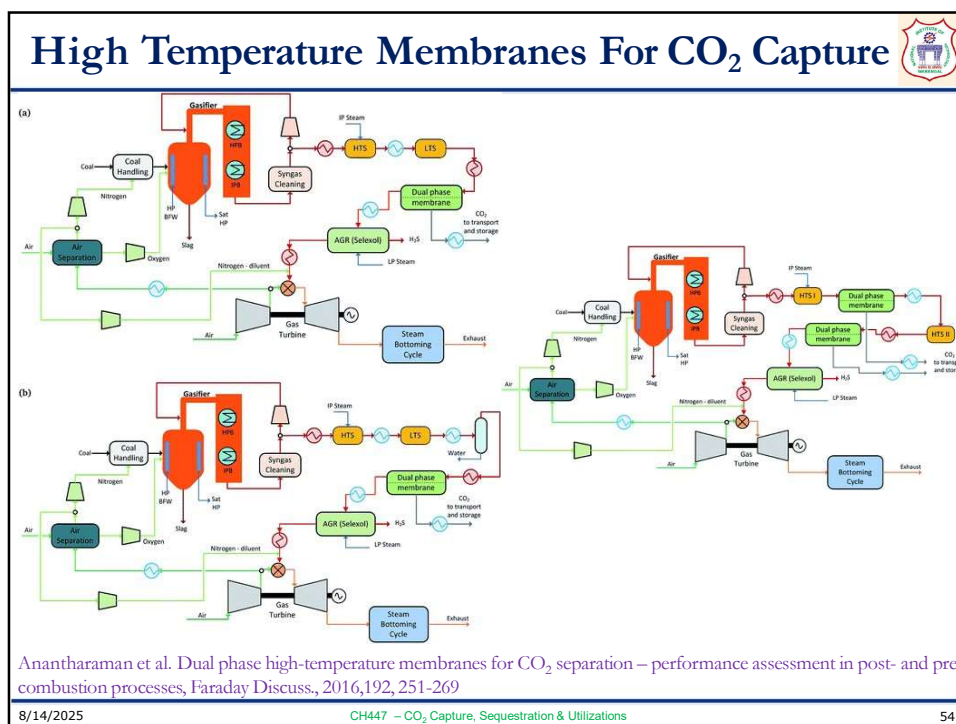
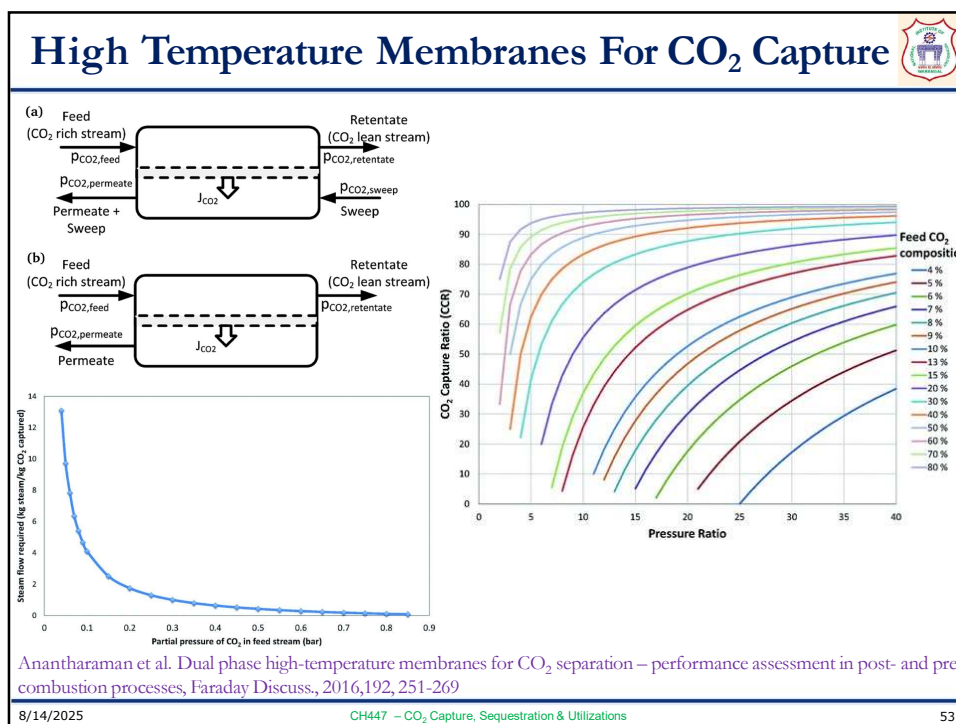
× encapsulation attempted and failed

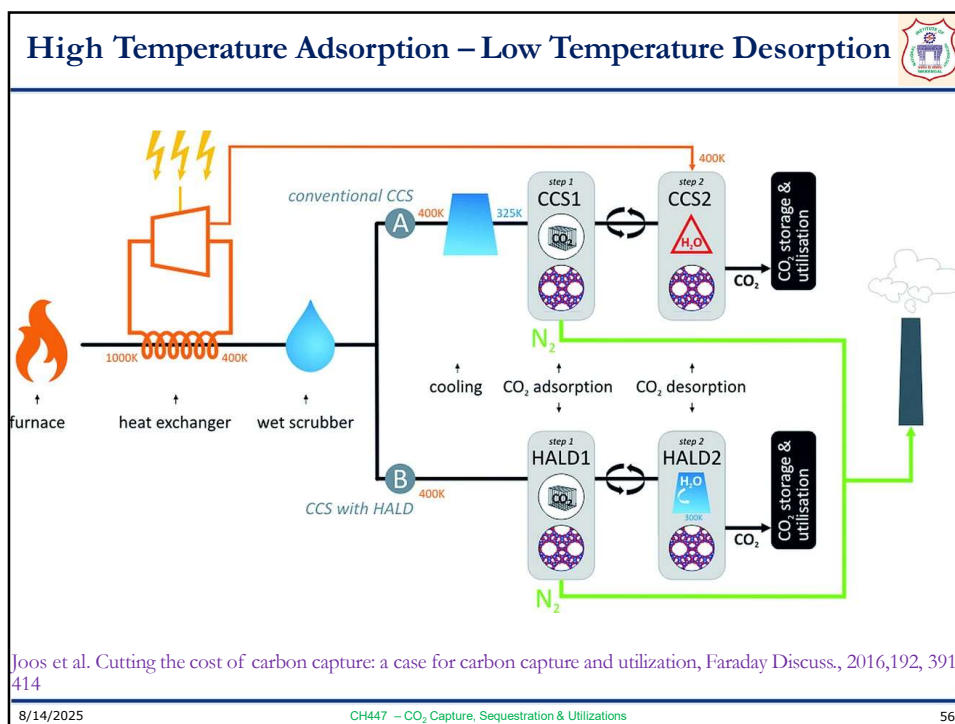
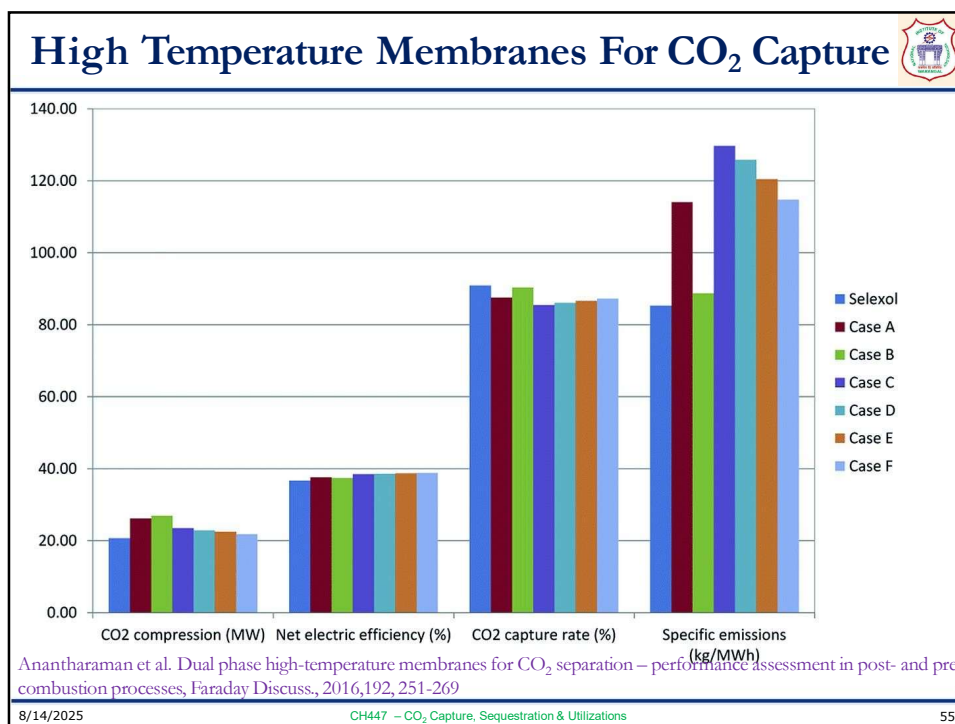
✓ encapsulation succeeded

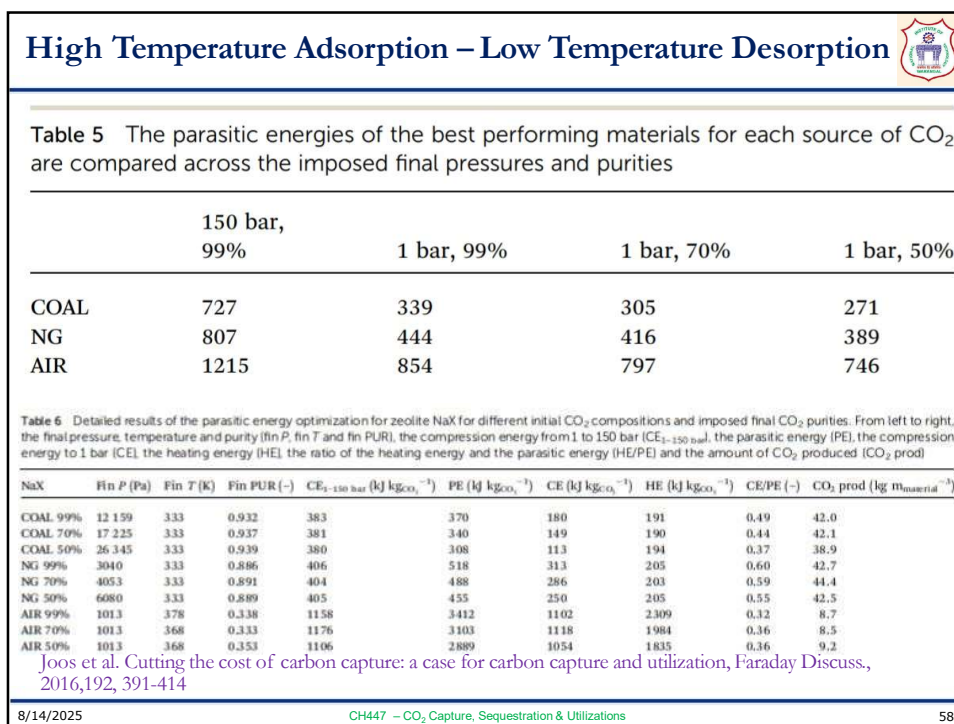
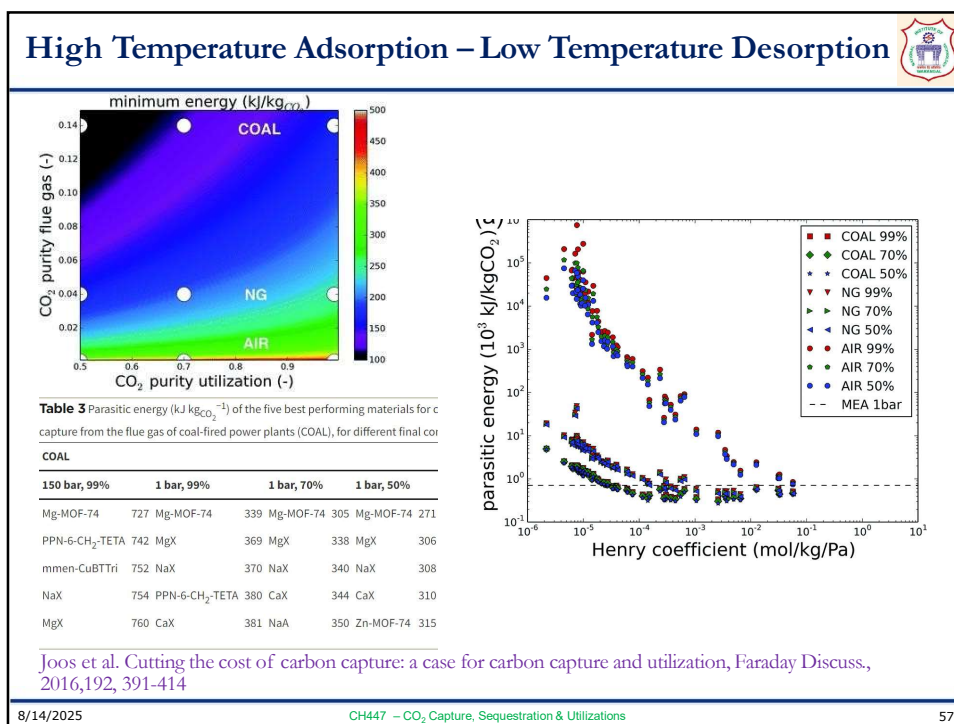
Stolaroff et al, Microencapsulation of advanced solvents for carbon capture, Faraday Discuss., 2016,192, 271-281

8/14/2025 CH447 – CO₂ Capture, Sequestration & Utilizations 51









References



1. Peter Styring, Elsje Alessandra Quadrelli, Katy Armstrong, Carbon dioxide utilization: Closing the Carbon Cycle, Elsevier, 2015.
2. Goel M, Sudhakar M, Shahi RV, Carbon Capture, Storage and, Utilization: A Possible Climate Change Solution for Energy Industry, TERI, Energy and Resources Institute, 2015.
3. Amitava Bandyopadhyay, Carbon Capture and Storage, CO₂ Management Technologies, CRC Press, 2014. 4.
4. Fennell P, Anthony B, Calcium and Chemical Looping Technology for Power Generation and Carbon Dioxide (CO₂) Capture, Woodhead Publishing Series in Energy: No. 82, 2015.
5. Mercedes Maroto-Valer M, Developments in Innovation in Carbon Dioxide Capture and Storage Technology: Carbon Dioxide Storage and Utilization, Vol 2, Woodhead Publishing Series in Energy, 2014.